

Commercial Applications of Nanotechnology

The scope of application of nanotechnology is very wide. Advanced research organisations around the world have identified five key market areas where nanotechnology can bring market-changing competitiveness as detailed in this article

RATHINDRA NATH BISWAS

1 Photovoltaic cells

In order to alleviate global warming by reducing greenhouse gas emission from combustion of fossil fuel, various countries are committed to curb 40-45 per cent carbon emission and increase the share of renewable energy to about 15 per cent of the total energy generated by 2020. This presents a big market opportunity for photovoltaic (PV) manufacturers.

More than 80 per cent of PV modules installed worldwide are made of crystalline silicon (c-Si). One of the main drawbacks of c-Si modules is their high cost due to use of expensive raw material, complicated manufacturing process and sophisticated equipment. Hence thin-film (TF) PV modules are rapidly gaining market shares because of their lower cost per watt, especially due to much lower use of the critical light-absorbing materials.

Thin-film PV cells are made from either amorphous silicon (a-Si) or copper indium gallium diselenide (CIGS) and organic photovoltaic. The research focuses on the key to fabricating low-cost thin-film solar cells, which includes controlling the microstructure, morphology and composition of their thin films and their interfaces.

Of all the thin-film solar cell technologies, CIGS is the most promising due to its higher conversion efficiency, lower toxicity and lower cost of manufacturing than a-Si and Cd-Te cells. Reliable control of CIGS thin-film morphology and its composition has been developed, which increases its efficiency. The process involves use of selenium powder by evaporation instead of using highly toxic H_2Se gas as the source of selenium. Large-area purge flow control evaporation chamber has been designed in conjunction with advanced rapid thermal process and precise chemical bath deposition of barrier layer for large area uniformity control. All these steps yielded a better CIGS solar cell.

Although the conversion efficiency of organic photovoltaic cell is still low, it has a high potential to bring affordable solar energy to different markets as it involves simple processing and equipment. High-purity phenyl-C60-butyric acid (PCBM) and low-cost novel donor materials are being evaluated. Process optimisation concentrates on controlling the microstructure morphology and composition for

maximising the conversion efficiency. Low-cost manufacturing is possible by adopting roll-to-roll manufacturing line.

Light management significantly increases the efficiency of PV cells, especially in thin-film solar cells that are only a few micrometres thick. One of the approaches is the use of a photonic crystal for selective spectrum tailored for different bandgaps of solar cells. Photonic crystal is a macroscopic, periodic dielectric structure that possesses spectral gaps (stop bands) for electromagnetic waves, in analogy with the energy bands and gaps in regular semiconductors. The advantages of using photonic crystals for light redirection compared to organic dyed solar cells are lower cost, higher efficiency and better directional performance.

As shown in Fig. 1, a photonic crystal film of about $10\mu m$ thickness can be coated on a 3.2mm thick glass substrate which can laterally redirect light in different spectra. Different combinations of spectra can be made by suitably designing the photonic crystal. As the coating is thin enough, a transparent solar panel can be realised with a proper design where solar cells can be put on edges of the glass for electricity generation.

In a conventional Si PV cell, a lot of light energy is lost due to reflection. Conventional multistack dielectric antireflection coating may be used, which is costly and shows only limited bandwidth.

A maskless process requiring just a few steps has been developed to provide antireflectivity on silicon, which has low reflectivity over a wide acceptance angle. At a wide bandwidth of 400-800 nm, weighted reflectivity on silicon can be substantially reduced from 41 to 3 per cent at an acceptable angle of 15° and from 43 to 15 per cent at an acceptance angle of 60° . Fig. 2 shows such surface morphology on silicon with nanopillars providing low reflectivity over a wide acceptance angle.

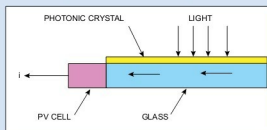


Fig. 1: PV cell using photonic crystal

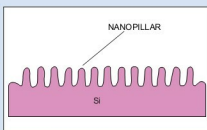


Fig. 2: PV cell with nanopillar